

L-14 Notes on L_p Spaces

L_p Space Definition

Given a probability space $(\Omega, \mathcal{F}, P)$.

Definition: A set $L_p = \{X : \Omega \rightarrow \mathbb{R} \mid \mathbb{E}[|X|^p] < \infty\}$ is called the L_p space associated with $(\Omega, \mathcal{F}, P)$.

Definition: A sequence of random variables $\{X_n\}_{n=1}^\infty$ from L_p we say " X_n converges to X in L_p ", and write as $X_n \xrightarrow{L_p} X$, if $\mathbb{E}[|X_n - X|^p] \rightarrow 0$ as $n \rightarrow \infty$.

Vector Space

A set $V \neq \phi$ with two operations:

1. **Vector Addition:** $+: V \times V \rightarrow V$, such that $(\underline{x}, \underline{y}) \mapsto \underline{x} + \underline{y}$. For $V = \mathbb{R}^d$,

$$\text{if } \underline{x} = \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_d \end{pmatrix}, \underline{y} = \begin{pmatrix} y_1 \\ y_2 \\ \vdots \\ y_d \end{pmatrix}, \text{ then } \underline{x} + \underline{y} = \begin{pmatrix} x_1 + y_1 \\ x_2 + y_2 \\ \vdots \\ x_d + y_d \end{pmatrix}.$$

2. **Scalar Multiplication:** $\cdot : \mathbb{R} \times V \rightarrow V$, such that $(c, \underline{v}) \mapsto c \cdot \underline{v}$. For

$$V = \mathbb{R}^d, \text{ if } c \in \mathbb{R}, \text{ then } c \cdot \underline{x} = \begin{pmatrix} cx_1 \\ cx_2 \\ \vdots \\ cx_d \end{pmatrix}.$$

The pair $(V, +)$ has the following properties:

- Associativity
- Commutativity
- Identity element: $\exists \underline{0} \in V$ s.t. $\underline{v} + \underline{0} = \underline{0} + \underline{v} = \underline{v}$
- Inverse element: $\forall \underline{v} \in V, \exists \underline{u} \in V$ such that $\underline{v} + \underline{u} = \underline{u} + \underline{v} = \underline{0}$. Such a $\underline{u}$ will often be denoted by $(-\underline{v})$. Together these properties make $(V, +)$ a commutative group.

Properties of scalar multiplication:

- **Associativity:** $c_1(c_2 \cdot \underline{v}) = (c_1 c_2) \underline{v}$
- **Distributivity:** $c \cdot (\underline{u} + \underline{v}) = c \cdot \underline{u} + c \cdot \underline{v}$
- **Identity:** $1 \cdot \underline{v} = \underline{v}$, where $1 \in \mathbb{R}$

The zero vector and the inverse vector in $\mathbb{R}^d$ are:

$$\underline{0} = \begin{pmatrix} 0 \\ 0 \\ \vdots \\ 0 \end{pmatrix} \quad -\underline{x} = \begin{pmatrix} -x_1 \\ -x_2 \\ \vdots \\ -x_d \end{pmatrix}$$

Theorem: Let $p \geq 1$. The space L_p is a vector space under the usual addition of random variables and multiplication of a random variable and real number.

Given X, Y are random variables and $c \in \mathbb{R}$:

- Point by point addition: $(X + Y) : \Omega \rightarrow \mathbb{R}$ is defined by $\omega \mapsto X(\omega) + Y(\omega)$.
- Scalar multiplication: $(c \cdot X) : \Omega \rightarrow \mathbb{R}$ is defined by $\omega \mapsto c \cdot X(\omega)$.

To show that L_p is a vector space, we must show closure under addition and scalar multiplication i.e.

1. If $X, Y \in L_p$, then $X + Y \in L_p$.
2. If $c \in \mathbb{R}$ and $X \in L_p$, then $c \cdot X \in L_p$.

Also, note that $L_p \neq \emptyset$ because the zero random variable $X \equiv 0$ is in L_p , as $\mathbb{E}[|0|^p] = 0 < \infty$.

If $X, Y \in L_p$, it means $\mathbb{E}[|X|^p] < \infty$ and $\mathbb{E}[|Y|^p] < \infty$. We need to show $\mathbb{E}[|X + Y|^p] < \infty$.

Consider the case when $p = 1$ (L_1):

Given $\mathbb{E}[|X|] < \infty$ & $\mathbb{E}[|Y|] < \infty$. We must show $\mathbb{E}[|X + Y|] < \infty$. The triangle inequality will do: $\mathbb{E}[|X + Y|] \leq \mathbb{E}[|X| + |Y|] = \mathbb{E}[|X|] + \mathbb{E}[|Y|] < \infty$.

Consider the case when $p = 2$ (L_2):

Given $\mathbb{E}[|X|^2] < \infty, \mathbb{E}[|Y|^2] < \infty$. We must show $\mathbb{E}[(X + Y)^2] < \infty$.

Using the Cauchy-Schwarz inequality: $|E[XY]| \leq \sqrt{E[X^2]E[Y^2]} < \infty$.

$$\mathbb{E}[(X + Y)^2] = \mathbb{E}[X^2 + 2XY + Y^2] = \mathbb{E}[X^2] + 2\mathbb{E}[XY] + \mathbb{E}[Y^2] < \infty$$

For general $p \geq 1$:

Consider $a, b \geq 0$. We can establish the inequality $(a + b)^p \leq 2^p(a^p + b^p)$. Since $(a + b) \leq 2 \max(a, b)$, we have $(a + b)^p \leq (2 \max(a, b))^p = 2^p \max(a^p, b^p) \leq 2^p(a^p + b^p)$.

For general $p \geq 1$, consider

$$X, Y \geq 0$$

Step 1:

$$(X + Y)^p \leq 2^p (X^p + Y^p)$$

$$E[(X + Y)^p] \leq 2^p (E[X^p] + E[Y^p]) < \infty$$

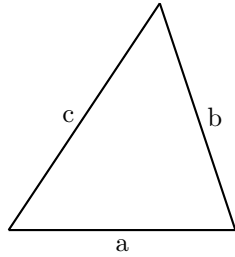
Step 2: General X, Y

$$E[|X + Y|^p] \leq E[|X| + |Y|]^p$$

$$\leq E[|X|^p] + E[|Y|^p] < \infty$$

Triangle Inequality:

$$|a + b| \leq |a| + |b|$$



Closure under scalar multiplication:

Let $c \in \mathbb{R}$ and $X \in L_p$.

$$\mathbb{E}[|c \cdot X|^p] = \mathbb{E}[|c|^p |X|^p] = |c|^p \mathbb{E}[|X|^p] < \infty$$

This shows $c \cdot X \in L_p$. If

$$E[|c \cdot x|^p] = E[|c|^p |x|^p] = |c|^p E[|x|^p] < \infty \quad \square$$

$$\Omega = \{A\}$$

$$X : \Omega \rightarrow \mathbb{R}$$

$$X(A) \in \mathbb{R}$$

Set of r.v.'s is same as $\mathbb{R}$.

$$L_p = \text{all r.v.'s on } \Omega.$$

The set of random variables on Ω , and thus the space L_p , can be very large.

- If $\Omega = \{H, T\}$, a random variable $X : \Omega \rightarrow \mathbb{R}$ is defined by the pair $(X(H), X(T))$. The set of all such random variables is isomorphic to $\mathbb{R}^2$. So L_p is isomorphic to $\mathbb{R}^2$.

- Ω is 'very large'.

$X : \Omega \rightarrow \mathbb{R}$ plenty of r.v.s.

L_p may be having many very different r.v.s.

$$f : \Omega \rightarrow S$$

$$S^\Omega$$

 L_p Norm

For any $X \in L_p$, we can define a function, often written as $\|\cdot\|_p$:

$$\|X\|_p = (\mathbb{E}[|X|^p])^{1/p}$$

The function $\|\cdot\|_p : L_p \rightarrow [0, \infty)$ is defined as $X \mapsto \|X\|_p = (\mathbb{E}[|X|^p])^{1/p}$.

In particular:

- For $p = 1$, $\|X\|_1 = \mathbb{E}[|X|]$
- For $p = 2$, $\|X\|_2 = \sqrt{\mathbb{E}[X^2]}$

Note: If $\mathbb{E}[X] = 0$, then $\|X\|_2$ is the standard deviation of X , $SD(X)$.

Properties of a Norm

In particular

For $p = 1$:

$$\|X\|_1 = E[|X|]$$

For $p = 2$:

$$\|X\|_2 = \sqrt{E[X^2]}$$

Note: if $E[X] = 0$ then $\|X\|_2 = SE(X)$.

NORM

1. Non-negative
2. Follows Triangle inequality
3. Norm zero iff vector is null vector.
4. $\|c \cdot v\| = |c| \|v\|$
1. For $c \in \mathbb{R}$, $X \in L_p$:

$$\begin{aligned} \|c \cdot X\|_p &= (E[|c \cdot X|^p])^{\frac{1}{p}} \\ &= (|c|^p E[|X|^p])^{\frac{1}{p}} \\ &= |c| \cdot \|X\|_p \end{aligned}$$

2. If $\|X\|_p = 0$:

$$\begin{aligned} \|X\|_p = 0 &\implies (E[|X|^p])^{\frac{1}{p}} = 0 \\ &\iff E[|X|^p] = 0, \text{ a.s.} \\ &\iff X = 0, \text{ a.s.} \end{aligned}$$

The NORM requirement 3 fails. :(

L-14 ends